Multiple Linear Regression

Diagnostics

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MLR … diagnostics

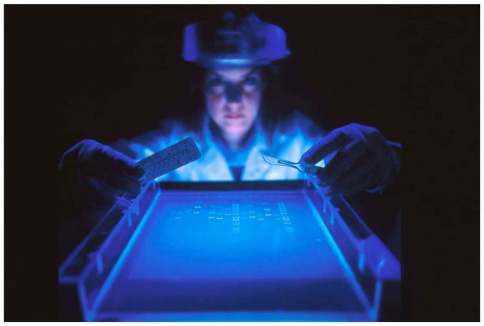

Photo by National Cancer Institute on Unsplash

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continuous

$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + e$

predictors, covariates, independent variables

$X_2 = \begin{cases} 0 \\ 1 \\ 2 \end{cases}$ → 3 categories

Last class

Multiple Linear Regression (MLR)

R_i factor(X_2)

as.factor(X_2)

→ 2 estimates for β_{2i}: $\hat{\beta}_{21}$, $\hat{\beta}_{22}$

- Categorical input variables with 2 or more levels
- Additive MLR: with different type of input variables → continuous or categorical
- MLR with interaction terms: interactions between continuous and categorical input variables (or continuous)

$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + e$

$H_0: \beta_3 = 0$

$H_1: \beta_3 \neq 0$

- Additive MLR with any type of input variables

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Today

Assumptions and Diagnostics

- What assumptions are you making when fitting a linear model?
- How do we check the validity of these assumptions?
- What happens if these assumptions are violated?

$y = f(x_1, x_2) + e$

All models need assumptions

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$

linear additive

distributional assumption

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1. Assumptions

L. **Linear relation** between response and predictors;

I. Errors are **Independent**; $e_1, e_2, \cdots, e_n$

N. Conditional distribution of the error terms is **Normal** (thus that of the response if errors are iid)

E. The variance of the error terms are **Equal**

- linear
- independence
- normality
- constant variance

These assumptions are needed at different stages of the analysis (some are *more "needed"* than others).

Note that these are **assumptions** and do not necessarily hold true. We should consider how to check their validity.

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1.1 Assumption 1: Linear Relation

So far, we have been using a line to model the association between the predictors and the response. Not surprisingly, the relationship should be close to a line for the model to be reliable. If the relationship is highly non-linear, then our model is dubious.

Two questions immediately arise:

1. Diagnosis: how do we detect non-linearity?

2. What can we do about it?

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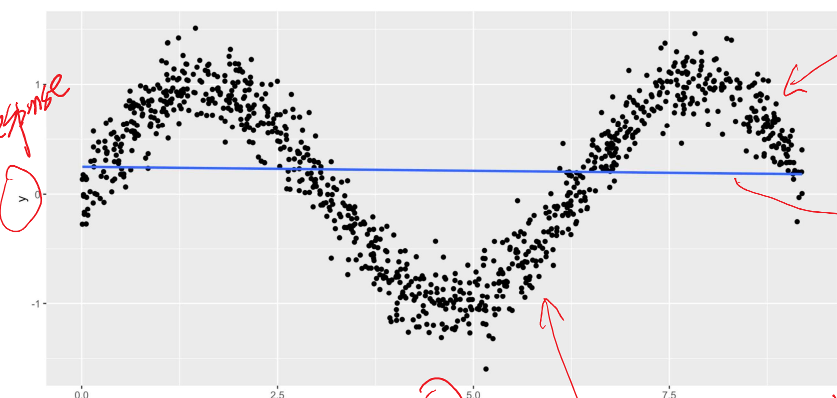
Very dubious model

data

response



fitted linear model

one predictor

linear assumption is very wrong!

Q: What if we have two or more predictors?

A: We need other methods.

1.1.1 Diagnosis: Linearity

Keep in mind that residuals hold all the data that the model does not account for. We can examine the residuals to understand more about our model.

One useful way to identify any (non-linear) patterns in the data that the model doesn't capture is by plotting the fitted values against the residuals.

After model fitting, we have $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$

Fitted values (responses): $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i}$, (no e_i)

Residuals: $r_i = y_i - \hat{y}_i$, $i = 1, 2, \cdots, n$

model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$

Data: ~~(x_1, y_1)~~, (x_{11}, x_{21}, y_1), (x_{12}, x_{22}, y_2), $\cdots$, (x_{1n}, x_{2n}, y_n)

model: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + e_i$, $i = 1, 2, \cdots, n$

observed response

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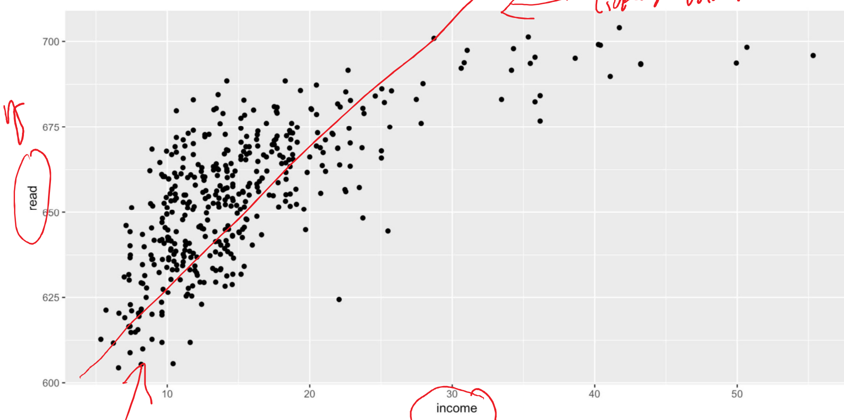
Example: reading score vs income



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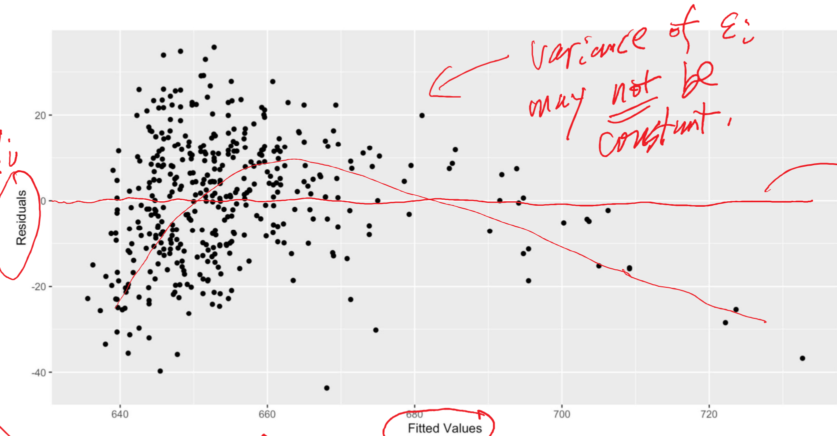
Looking at the residuals

model: $y=\beta_0+\beta_1 x+e$

☆ residual plot



line $Y=0$

If the model fits data well, the residuals should ~~all~~ be around line $Y=0$.

$r_i = y_i - \hat{y}_i$

Some patterns

Otherwise, something may be wrong with the model.

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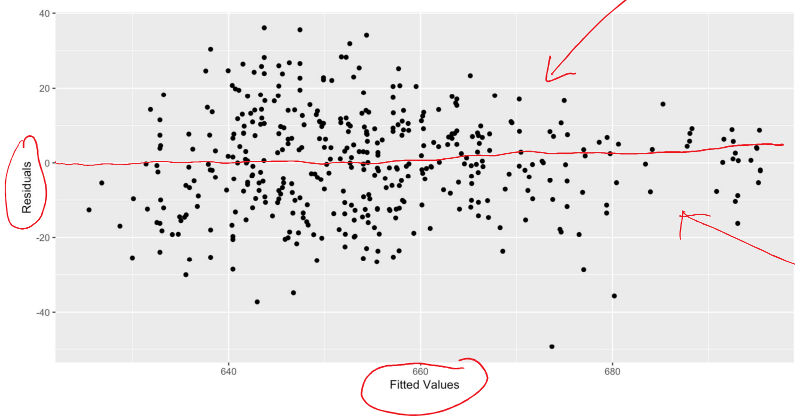
constant variance assumption

Adding a quadratic term:

```
caschools_quad <- lm(read ~ income + I(income^2), data = caschools)
```



Constant variance assumption may be reasonable.

revised model

$y = \beta_0 + \beta_1 x + \beta_2 x^2 + e$

revised model fits the data better

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1.1.2 The misleading word `Linear`

Although we have only been fitting lines so far, a linear regression can have any covariate in the model. For example,

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_1 X_{2i} + \varepsilon_i$$

The covariates X_1 and X_2 can be anything, such as, $X_1 = \text{income}$ and $X_2 = \text{income}^2$, which results in

$$\text{read}_i = \beta_0 + \beta_1 \times \text{income}_{1i} + \beta_1 \times \text{income}_{2i}^2 + \varepsilon_i$$

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Linear

L. Linear relation between response and predictors;

The term "linear" refers to the relation between the predictors and the response, aka "linear in the parameters". → not in predictors

- The least square method will work precisely the same way, even if some of the predictors are transformations of other variables, e.g., $\log(X)$, X^2, $\sqrt{X}$
- Just be careful, as this will affect the interpretation of the model.

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1.1.3 What can we do about non-linearity

e.g. x^2, $\log(x)$

We can include transformations of the covariates and interaction terms between covariates to make the model more "powerful". But remember that this this will affect the interpretation of the model.

⇒ then, check residual plots to see if they look better.

Note that the relation between a linear regression may not be a good fit for the data, regardless of transformation considered. In such cases, you may want to consider using different models or algorithms.

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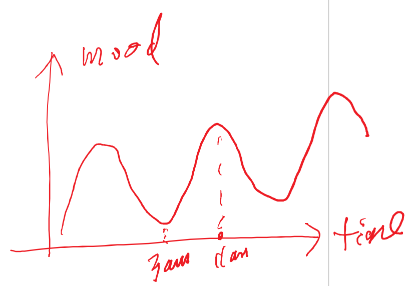
1.2 Assumption 2: Independence

We assume we have a random sample. The independence of the errors can usually be assessed from the study's design.

For example, we can safely assume that the scores from one district do not affect the scores from another district.



Commonly data that violate independence:

- temporal data (data collected over time)
- multiple measurements of the same subject → e.g., your homework scores are not independent.

longitudinal data
repeated measurements

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1.2.1 Diagnosis: Independence

This is tricky. The correlation can be very different in different problems. However, a residual plot might be helpful.

There are also statistical tests for some types of residual correlation (e.g., Durbin-Watson test). However, these tests are outside the scope of this course.

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Examples: independence violation

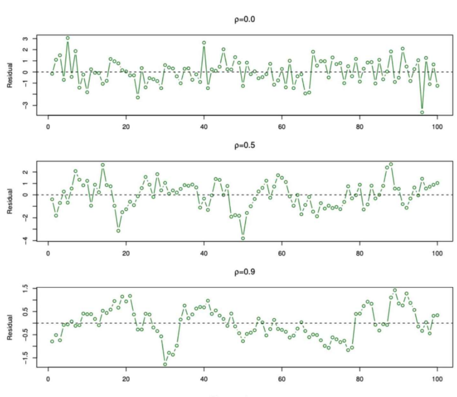

FIGURE 3.10. *Plots of residuals from simulated time series data sets generated with differing levels of correlation ρ between error terms for adjacent time points.*

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1.2.2 Violation of independence

The biggest issue here is that the std. errors of our estimators $\hat{\beta}$ s are biased; therefore, the confidence intervals and hypothesis tests we learned are invalid.

1.2.3 What can we do about it?

Some variations of Least Squares have been proposed to address with this problem, but they have some limitions.

There are also other methods appropriate for correlated observations (e.g., times series analysis, longitudinal data analysis) but beyond the scope of this course.

do not use linear regression models

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1.3 Assumption 3: Constant Variance of the Errors (homoscedasticity)

Remember from the model equation:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad , i=1,2,\ldots,n$$

$\varepsilon_i \sim N(0, \sigma^2)$

where ε_i are random components for sample i.

One of the assumptions is that each one of these ε_i have the same standard deviation (no subscript i)

$$Var(\varepsilon_i) = \sigma^2$$

a constant, i.e, σ^2 does not depend on i.

$\parallel$

$Var(y_i)$

check this based on residual plot.

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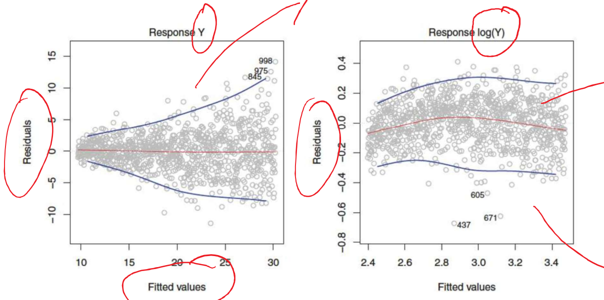
1.3.1 Diagnosis: heteroschedasticity

$f(x) = e^x$

Once again, the fitted vs residuals plot is helpful in detecting heteroscedasticity.

variance not constant



constant variance assumption is more reasonable.

model based on $\log(y)$ fits data better.

FIGURE 3.11. *Residual plots. In each plot, the red line is a smooth fit to the residuals, intended to make it easier to identify a trend. The blue lines track the outer quantiles of the residuals, and emphasize patterns.* Left: *The funnel shape indicates heteroscedasticity.* Right: *The response has been log transformed, and there is now no evidence of heteroscedasticity.*

You should see a consistent variation on the residual plot as the value of the fitted variable changes.

1.3.2 Violation of homoscedasticity

Once again, this affects the standard error of our estimators. Consequently, the confidence intervals and p-values are no longer valid.

$\log(y) = \beta_0 + \beta_1 X + e$

$y = e^{\beta_0 + \beta_1 X} \times e$

1.3.3 What can we do about it?

Variance stabilizing transformation of the response (e.g., $\sqrt{Y}$, or $\log(Y)$).

Another possibility is to use *Weighted Least Square* when we have an idea of how the variance changes with the response (out of the scope of this course).

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1.5 Assumption 4: Normality

The violation of this assumption is (arguably)less severe than the others.

- The errors do not necessarily need to be Normal to have valid inference results
  - If the sample size (n) is large, the CLT gives approximations for the sampling distribution
  - Bootstrapping can also be used to approximate the sampling distribution

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Why Normality?

However, if the conditional distribution of the errors is Normal, it can be proved that the conditional expectation of the response is linear! So our model is good!!

without this assumption, it's hard to compute a p-value

- Q-Q plots and histograms of the residuals can be used to diagnose this problem
- Variables transformations can be used as possible "remedies"
- Most importantly, we should think if we can rely on the CLT or if bootstrapping is preferred

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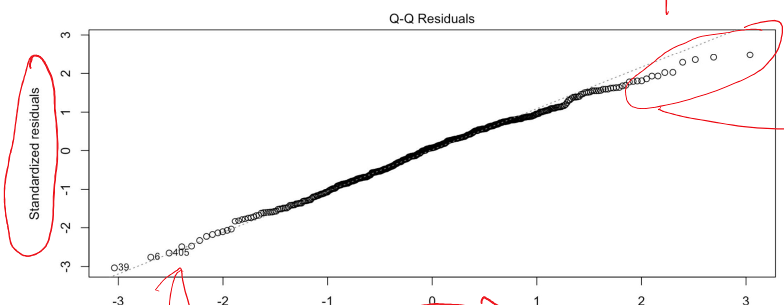
1.6 R plots everything for you

```
plot(caschools_lm, 2)
```

Q-Q plot to check normal assumption



based on N(0,1)

If normality holds, the points should roughly on a straight line.

Good

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2. Multicollinearity

Multicollinearity occurs when there is a strong association between two or more covariates. These covariates bring similar information to the model, and we have trouble isolating their effects.

For example, consider the following (drastic) model:

$$\text{read}_i = \beta_0 + \beta_1 \times \underbrace{\text{income}}_{X_1} + \beta_2 \times \underbrace{\text{income}*1000}_{X_2} + \varepsilon_i$$

In other words, we are using `income` in 1000USD and in USD to explain `read`.

$Cor(X_1, X_2) \approx 1$


```
caschools2 <- caschools %>% mutate(incomeUSD =income*1000)

ill_model <- lm(read ~ income + incomeUSD, data = caschools2)

tidy(ill_model)
```

```
# A tibble: 3 × 5
  term        estimate std.error statistic   p.value
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)   625.      1.65       379.   0
2 income          1.94    0.0975      19.9  1.46e-62
3 incomeUSD      NA      NA           NA   NA
```

In this drastic case, R could not even estimate the coefficients since the variable `incomeUSD` is essentially the same as `income`; one is just a linear transformation of the other.

The consequence is that R cannot separate the effect of these variables (of course, since they are the same).

Where is the problem?

Now, let's delve into the technical aspect that caused R to struggle. Remember, in the least square, our goal is to minimize the Square Sum of Residuals (SSR). The SSR in R's output for our `model_example` is:

$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

```
sum(ill_model$residuals**2)
```

```
[1] 86914.13
```

This model corresponds to:

$$\text{read}_i = 625 + 1.94 \times \text{income}_i$$

But many different models (in fact, infinite) that would give us exactly the same SSR, for example:

Model 2:

$$\text{read}_i = 625 + 0.97 \times \text{income}_i + 0.00097 \times \text{incomeUSD}_i$$

Model 3:

$$\text{read}_i = 625 + 0.485 \times \text{income}_i + 0.001455 \times \text{incomeUSD}_i$$

Model 4:

$$\text{read}_i = 625 + 0.00194 \times \text{incomeUSD}_i$$

```r
beta0 <- 625
beta1 <- 1.94


caschools2 %>%
    mutate(pred_model_1 = beta0 + (beta1) * income,
           pred_model_2 = beta0 + (beta1/2) * income + beta1/(2*1000)*incom
           pred_model_3 = beta0 + (beta1/4) * income + beta1*3/(4*1000)*inc
           pred_model_4 = beta0 + (beta1 /1000) *incomeUSD,
           red_model_1 = read - pred_model_1,
           red_model_2 = read - pred_model_2,
           red_model_3 = read - pred_model_3,
           red_model_4 = read - pred_model_4) %>%
    summarise(SSR_model_1 = sum(red_model_1**2),
              SSR_model_2 = sum(red_model_2**2),
              SSR_model_3 = sum(red_model_3**2),
              SSR_model_4 = sum(red_model_4**2))
```

```
  SSR_model_1 SSR_model_2 SSR_model_3 SSR_model_4
1    86941.73    86941.73    86941.73    86941.73
```

Since `incomeUSD = 1000*income`, we can shift "weight" from one variable to another. All models are equivalent.

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2.1 Consequences of multicollinearity

- Multicollinearity inflates the standard errors of our estimates, making our confidence intervals wider than necessary and making it harder to reject the null-hypothesis of the coefficients.

inflates p-values

$$Z = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$$

- In practice, we don't observe perfect multicollinearity.

- Also, the association doesn't need to be between two variables alone (a common mistake of students).

Think of multicollinearity as if we regress one of the covariates as a response on all other covariates and obtain a very good

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between

2.2 Diagnosis: multicollinearity

correlation between each pair of covariates (predictors)

- A very useful way is to check the pairwise correlation between all the covariates. If one of the covariates is highly correlated with another, you should pay attention. However, **this is not enough**, as multicollinearity can happen when considering more than two covariates.

Continuous

- Another useful way is to use the VIF. See Section 3.3 of ISL.

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VIF

The VIF checks the increase in the standard error of the coefficients when fitted alone compared to when fitted with all other variables.

The book suggests a VIF higher than 5 or 10 suggests multicollinearity might be problematic. **This is a guideline!!**

Example: Pinguins

Let's fit a MLR to the penguins dataset using all the variables except year (which is the same for all rows).

```
penguins_clean <-
    penguins %>%
    drop_na()

cor(penguins_clean %>%
    select(flipper_length_mm, bill_length_mm, bill_depth_mm)) %>%
    round(2)
```

cor(X1, X2, X3)

X1 X2 X3

```
                  flipper_length_mm bill_length_mm bill_depth_mm
flipper_length_mm              1.00           0.65         -0.58
bill_length_mm                 0.65           1.00         -0.23
bill_depth_mm                 -0.58          -0.23          1.00
```

As we can see what have somewhat high correlation. So we should stay alert for multicollinearity.

cor(X1, X3) cor(X1, X2)

$\rightarrow GVIF^{\frac{1}{2 \cdot df}}$ df = # of predictors - 1

	GVIF	Df	GVIF^(1/(2*Df))
X_1 species	63.52	2	2.82
X_2 island	3.73	2	1.39
X_3 bill_length_mm	6.10	1	2.47
X_4 bill_depth_mm	6.10	1	2.47
X_5 flipper_length_mm	6.80	1	2.61
X_6 sex	2.33	1	1.53

→ compare with $\sqrt{5}$ or $\sqrt{10}$

$\sqrt{5} = 2.23$

There may be multicollinearity problem

This function actually returns the generalized VIF as well.

The **generalized VIF** (right column) is preferred because the VIF of categorical covariates are affected by the number of categories they have. GVIF, in the right column, accounts for that.

You can either take the square of the right column and compare it with the guideline of 5 or 10 or compare the right column against $\sqrt{5}$ or $\sqrt{10}$.

2.3 What can we do about it?

The most straightforward approach is to drop the collinear variable.

Another possibility is to combine collinear variables. This might affect the interpretation of the model depending on how the aggregation is done.

As you can see, species has the highest VIF here higher than $\sqrt{5} = 2.23$ but still less than $\sqrt{10} = 3.16$.

Let's try removing `species` from the model.

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```r
tidy(penguins_lm_large)
```

```
# A tibble: 9 × 5
  term              estimate std.error statistic  p.value
  <chr>                <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)         -1500.     576.      -2.61  9.61e- 3
2 speciesChinstrap     -260.      88.6     -2.94  3.52e- 3
3 speciesGentoo         988.     137.       7.20  4.30e-12
4 islandDream           -13.1     58.5     -0.224 8.23e- 1
5 islandTorgersen       -48.1     60.9     -0.789 4.31e- 1
6 bill_length_mm         18.2      7.14     2.55  1.13e- 2
7 bill_depth_mm          67.6     19.8      3.41  7.34e- 4
8 flipper_length_mm      16.2      2.94     5.52  6.80e- 8
9 sexmale               387.      48.1      8.04  1.66e-14
```

→ remove predictors "species" and "year".

```r
tidy(lm(body_mass_g ~ . - species - year, data = penguins_clean))
```

```
# A tibble: 7 × 5
  term              estimate std.error statistic  p.value
  <chr>                <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)         -2116.     605.      -3.50 5.31e- 4
2 islandDream          -313.      54.9     -5.71 2.48e- 8
3 islandTorgersen      -224.      65.4     -3.42 7.13e- 4
4 bill_length_mm          6.05     4.97     1.22 2.24e- 1
5 bill_depth_mm         -50.3     16.2     -3.10 2.10e- 3
6 flipper_length_mm      33.9      2.50    13.6  1.73e-33
```

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VIF again

As you can see, the fit of the coefficients related to `island` has improved considerably.

Let's check again:

```r
lm(body_mass_g ~ . - species - year, data = penguins_clean) %>%
    vif()
```

```
                      GVIF Df GVIF^(1/(2*Df))
island            2.412912  2        1.246337
bill_length_mm    2.310218  1        1.519940
bill_depth_mm     3.198249  1        1.788365
flipper_length_mm 3.855613  1        1.963572
sex               1.978756  1        1.406683
```

$< \sqrt{5}$ after remove "species" and "year"

multicollinearity problem disappears.

Everything looks ok (for now)!